INKLOOM BRANDGUARD â ROUND 2 THE BUILD

Working Product Documentation, Feature Matrix & Technical Architecture

Project Submission Parameter	Evaluation Value
Event & Stage	CodeTitans India 2026 â Round 2 (The Build)
Team Name & Track	Team ALTAIR â Inkloom AI Brand Governance Platform
Team Leader	MD ANAS AKHTAR (AI/ML & Evaluation Systems Engineer)
Evaluation Scorecard	100 Points: Workflow (30), Tech Depth (25), Plan Progress (20), 3 Impro..
Build Status	Working End-to-End System Verified (Zero Sign-In Required)

Round 2 Executive Summary: Following selection as one of the top 300 teams out of 1,000+ applicants, Team ALTAIR has translated the Round 1 proposal into a fully functional, demonstrable product. Inkloom BrandGuard executes the complete pre-publication governance lifecycle: unapproved drafts are ingested, evaluated across 6 dimensions against the Brand Vault, flagged issues are explained with guideline citations, and context-preserving 1-click rewrites are applied under human control.

1. Core Workflow: End-to-End Execution

The core workflow runs seamlessly from raw draft ingestion to official compliance certification, completely fulfilling the central scoring requirement:

- â¢ 1. Multi-Format Draft Ingestion: Users can select from 4 pre-loaded real-world enterprise scenarios (Fintech Series B Press Release, Product Launch Blog, Social Ad Campaign, Clean Release) or paste custom markdown/text.
- â¢ 2. Real-Time 6-Dimension Inspection: The hybrid engine analyzes content in under 15ms across: (1) Tone & Voice; (2) Terminology & Glossaries; (3) Verified Factual Claims & SLAs; (4) Unsupported Promotional Guarantees; (5) Visual Brand Palettes; and (6) WCAG 2.1 Accessibility.
- â¢ 3. Assistive, Explainable Violation Cards: Each violation surfaces: (a) Dimension & Severity tag; (b) Flagged excerpt; (c) Exact guideline citation; (d) Business risk explanation; and (e) A 1-Click 'Accept & Apply' diff button.
- â¢ 4. Non-Destructive In-Editor Patching: Clicking 'Apply' patches the text in real-time, instantly recalculating the dynamic Brand Compliance Score and dimension health pills.
- â¢ 5. Role-Based Review & Certification: Collaborators toggle roles (Creator Alex -> Reviewer Elena -> Legal Marcus) to request changes, leave threaded comments, and grant approval. Approved assets trigger confetti and generate an exportable Brand Compliance Certificate.

2. List of Completed Features (Round 2 Deliverables)

Feature Module	Implemented Capabilities in Round 2 Build
Universal Content Studio	Live multi-format editor, word/char counters, readability scoring, and instant ..
Explainable Diff Engine	Guideline citations (e.g. CLAIM-01, TONE-01), severity badges, and 1-Click 'App..
Pre-Loaded Demo Scenarios	4 realistic enterprise test documents demonstrating subtle tone, SLA, and regul..
Brand Vault (Truth Source)	Interactive manager for Tone Guidelines, Terminology Glossaries (add/delete), a..
Review & Approval Hub	Status state machine (Draft -> Under Review -> Changes Requested -> Approved) a..
Governance Analytics	30-day recurring mistake breakdown (Promotional 42%, Terms 28%) and Guideline A..
Privacy & PII Redactor	Interactive client-side regex scrubber sanitizing emails, phone numbers, and co..
Audit Certificate Modal	Official printable compliance certificate with tamper-proof ID, audit dates, an..

3. Technology & Implementation Details

Inkloom BrandGuard was built from scratch during the hackathon using a modern, scalable engineering stack:

Architectural Component	Technology Used	Implementation Rationale
Frontend Framework	React 19 + TypeScript + Vite	Zero layout shift, instantaneous hot module rel..
Styling & Design System	Tailwind CSS + Lucide Icons	Enterprise dark-mode theme, accessible contrast..
Deterministic Engine	Regex & AST Pattern Matcher	Sub-15ms local execution for 500+ glossary term..
Semantic & AI Reasoning	Google Gemini 2.5 / 3.8 Flash API	Deep reasoning for tone subtlety, regulatory cl..
Local Privacy & Security	Client-Side Regex Scrubber	Zero leakage of confidential customer data, pre..
Vector Output	Pure Node.js PDF Stream Builder	Zero-dependency compiler for official submissio..

4. Explanation of Changes Made to the Original Plan

During development and testing, three strategic technical refinements were made to improve product reliability and enterprise real-world fit:

- â¢ Refinement 1 â Hybrid Local Routing vs. Blind LLM Streaming: Original Plan: Route every editor keystroke directly to an external LLM. Change: Implemented a sub-15ms local deterministic rule layer for exact terminology and banned casing, invoking Gemini only for complex semantic tone checks. This slashed latency by 90% and eliminated API cost limits.
- â¢ Refinement 2 â Addition of Guideline Ambiguity Index in Analytics: Original Plan: Basic chart showing violation counts. Change: Added an 'Ambiguity Heatmap' that measures which guidelines creators dispute or override most often (e.g. Rule CLAIM-01 at 34% override), transforming the tool into an active feedback loop for brand managers.
- â¢ Refinement 3 â In-Browser Client-Side PII Scrubbing: Original Plan: Server-side data processing. Change: Moved PII redaction directly into the client runtime before payload compilation to guarantee zero enterprise data leakage.

5. Three Major Improvements for Round 3 (The Finish)

In accordance with Round 3 criteria, Team ALTAIR has planned three substantial, meaningful functional additions:

- â¢ Improvement 1 â Multi-Modal Visual Hierarchy & Bounding-Box Scanner: Direct drag-and-drop audit of marketing banner graphics, slide decks, and Figma exports. Uses vision-LLM capabilities to verify logo clear-space padding (24px rule), visual hierarchy, and WCAG color contrast validation on raster assets.
- â¢ Improvement 2 â Multilingual & Regional Regulatory Compliance Engine: Extending the 6-dimension evaluation pipeline across 15+ global languages (French, German, Spanish, Japanese, Hindi) with regional advertising regulatory rules (EU AI Act, GDPR marketing disclosures, UK ASA guidelines).
- â¢ Improvement 3 â Enterprise CMS & Browser Extension Connectors: Headless publishing webhooks for WordPress, Webflow, and Contentful, paired with a Chrome/Edge browser extension that brings BrandGuard directly into Google Docs, Notion, and LinkedIn.

6. Team ALTAIR & Participant Contributions

Participant	Assigned Project Role	Specific Round 2 Contributions & Ownership
MD ANAS AKHTAR	Team Leader & AI/ML Engineer	6-dimension evaluation pipeline, deterministic/sema..
Collaborating Teammates	Full-Stack & UI/UX Engineers	React 19 Content Studio UI, visual diff engine, rol..